

MEDIATEK

Auto Camera ISP8S CDPHY Eye Scan User Guide

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Version History

Version	Date	Author	Description
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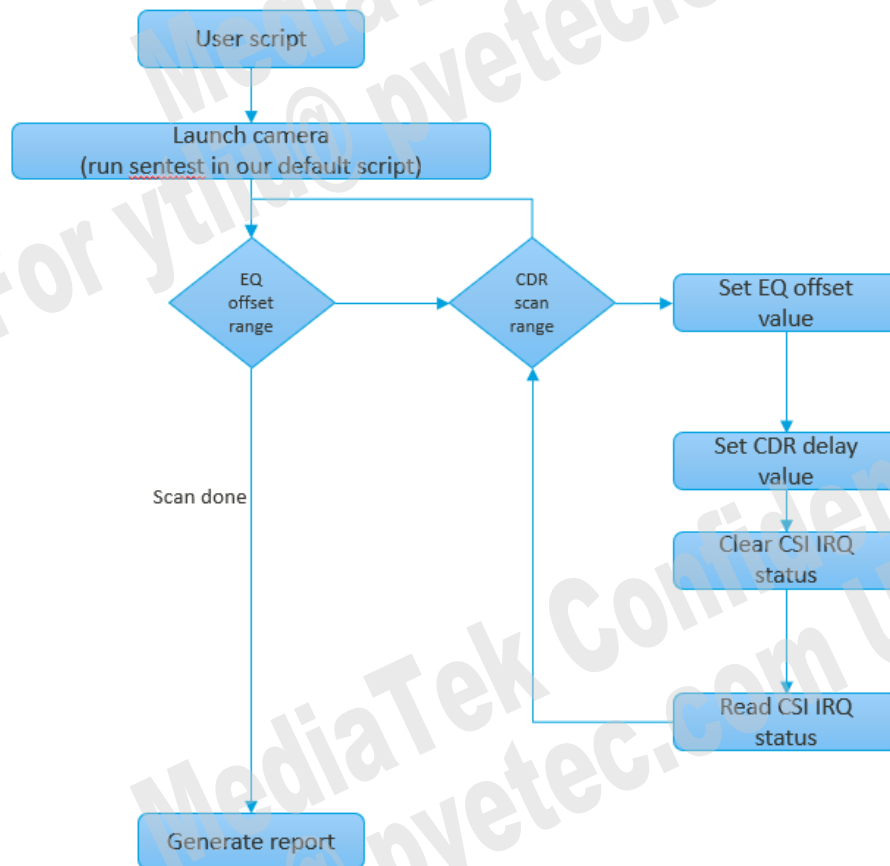
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1 Script Execution Flow

- Origin version

Note that the v.4 script would do “reboot” at the start and the end.



2 Environment Requirement

Android Debug Bridge (adb) & Python3 environment are required.

3 How To Run EYE_SCAN Script

3.1 File Path

The path of EYE_SCAN Python script is:

- Yocto:
src/multimedia/camera-hal/\${IC}/mtkcam-core/hw/sensor/tool/eye_scan/ eye_scan_tool.py
- Android:
vendor/mediatek/proprietary/hardware/mtkcam-core/hw/sensor/tool/eye_scan/eye_scan_tool.py

3.2 Parameter of Script

- Script arguments

Arguments	Default	Meaning
sensor_idx	1	Sensor index of SenTest 1: Main cam 2: Sub cam 4: Main2 cam 8: Sub2 cam 16: Main3 cam
sensor_mode	0	Sensor mode 0: Preview 1: Capture 2: Video ... (Pixel rate of each mode would impact scanning result.)
CDR_DELAY_start	0	Setting scan range of CDR_DELAY
CDR_DELAY_end	31	-
EQ_OFFSET_start	-31	Setting scan range of EQ_OFFSET
EQ_OFFSET_end	31	-
delay_time	0	Time duration (seconds) between flush IRQ status after new setting and check new IRQ status
serial	0123456789ABCDEF	Serial number of device (phone)

- Run CMD
 - Run with specified arguments

```
python eye_scan_tool.py --sensor_idx 1 --sensor_mode 1 --CDR_DELAY_start 0 --CDR_DELAY_end 31 --EQ_OFFSET_start -31 --EQ_OFFSET_end 31 --delay_time 0 --serial 0123456789ABCDE
```

- Run with default arguments

```
python eye_scan_tool.py
```

3.2.1 Modify Sensor_idx & Sensor_mode

- Command: `adb root & adb shell sentest`

```
C:\Users\MTK31366>adb shell sentest_v412
[main]sizeof long : 4
debug--[init_sensor_list] ioctl fail, line=300
[show_Sensors]sensorNum 5
[show_Sensors]name:SENSOR_DRVNAME_IMX089_MIPI_RAW type:0 max_scen: 33
[show_Sensors]index:0, SensorDevIdx:1
[show_Sensors]name:SENSOR_DRVNAME_S5K3P9SP_MIPI_RAW type:0 max_scen: 5
[show_Sensors]index:1, SensorDevIdx:2
[show_Sensors]name:SENSOR_DRVNAME_IMX598_MIPI_RAW type:0 max_scen: 8
[show_Sensors]index:2, SensorDevIdx:4
[show_Sensors]name:SENSOR_DRVNAME_IMX58_MIPI_RAW type:0 max_scen: 8
[show_Sensors]index:3, SensorDevIdx:16
[show_Sensors]name:SENSOR_DRVNAME_OV04B_MIPI_RAW type:0 max_scen: 12
[show_Sensors]index:4, SensorDevIdx:64
[main]Param: 1 <sensorDev> <scenario> <fps>
[main]<sensorDev> : testmdl(0), main(1), Sub(2), Main2(4), sub2(8), Main3(16)
[main]<scenario> : Pre(0), Cap(1), VD(2), slim(3), slim2(4), or TM pattern id
[main]<Test ID>: (total test case 0)
[main] ID: 000 Test_stream on
[main] ID: 001 Test_mini rate
```

- sensor_idx is set the same as SensorDevIdx.
- sensor_mode is set to what you want to scan (start from 0).

3.2.2 Modify C-PHY or D-PHY Range

- CDR_DELAY_start is set to 0 (iteration start from 0).
- C-PHY: CDR_DELAY_end is set to 31 (iteration range from CDR_DELAY_start to 31).
- D-PHY: CDR_DELAY_end is set to 254 (iteration range from CDR_DELAY_start to 254).
- CDR_OFFSET_start is set to -31 (iteration start from -31).
- CDR_OFFSET_start is set to 31 (iteration start from 31).

3.3 Result Check

You will get two csv files after scanning. In both files, x-axis --> CDR_DELAY, y-axis --> EQ_OFFSET.

- rg_val.csv:** Store all csi_irq_status values.
- result.csv:** Store all pass or not. For each cell, O means IRQ status is correct at such (CDR_DELAY, EQ_OFFSET) setting, while X means IRQ status is failed.

3.4 Example

- Terminal log
 - Origin version

```
PS C:\Users\mtk31366> python .\eye_scan_tool.py
adbd is already running as root
Start EYE_SCAN
sensor_idx=1 drv_sensor_idx=0
sensor mode = 0
CDR_DELAY from 0 to 31
EQ_OFFSET from -31 to 31
/sys/devices/platform/la00e000.seninf_top

----- launch camera -----

----- CDR scan -----

CDR_DELAY=0x0 EQ_OFFSET=-31 CSIRX_MAC_CS12_IRQ_STATUS=0x 2002000, [WRONG]
CDR_DELAY=0x1 EQ_OFFSET=-31 CSIRX_MAC_CS12_IRQ_STATUS=0x 2002000, [WRONG]
CDR_DELAY=0x2 EQ_OFFSET=-31 CSIRX_MAC_CS12_IRQ_STATUS=0x 2002000, [WRONG]
CDR_DELAY=0x3 EQ_OFFSET=-31 CSIRX_MAC_CS12_IRQ_STATUS=0x 3002000, [WRONG]
CDR_DELAY=0x4 EQ_OFFSET=-31 CSIRX_MAC_CS12_IRQ_STATUS=0x 7002012, [WRONG]
CDR_DELAY=0x5 EQ_OFFSET=-31 CSIRX_MAC_CS12_IRQ_STATUS=0x 7000012, [WRONG]
CDR_DELAY=0x6 EQ_OFFSET=-31 CSIRX_MAC_CS12_IRQ_STATUS=0x 700027a, [WRONG]
CDR_DELAY=0x7 EQ_OFFSET=-31 CSIRX_MAC_CS12_IRQ_STATUS=0x 7ff127b, [WRONG]
CDR_DELAY=0x8 EQ_OFFSET=-31 CSIRX_MAC_CS12_IRQ_STATUS=0x 700035a, [WRONG]
CDR_DELAY=0x9 EQ_OFFSET=-31 CSIRX_MAC_CS12_IRQ_STATUS=0x 7ff035b, [WRONG]
CDR_DELAY=0xa EQ_OFFSET=-31 CSIRX_MAC_CS12_IRQ_STATUS=0x 7ff135b, [WRONG]
CDR_DELAY=0xb EQ_OFFSET=-31 CSIRX_MAC_CS12_IRQ_STATUS=0x 700035c, [WRONG]
.....
CDR_DELAY=0x0 EQ_OFFSET=29 CSIRX_MAC_CS12_IRQ_STATUS=0x 2002000, [WRONG]
CDR_DELAY=0x1 EQ_OFFSET=29 CSIRX_MAC_CS12_IRQ_STATUS=0x 2002000, [WRONG]
CDR_DELAY=0x0 EQ_OFFSET=30 CSIRX_MAC_CS12_IRQ_STATUS=0x 2002000, [WRONG]
CDR_DELAY=0x1 EQ_OFFSET=30 CSIRX_MAC_CS12_IRQ_STATUS=0x 2002000, [WRONG]
CDR_DELAY=0x0 EQ_OFFSET=31 CSIRX_MAC_CS12_IRQ_STATUS=0x 0, [WRONG]
CDR_DELAY=0x1 EQ_OFFSET=31 CSIRX_MAC_CS12_IRQ_STATUS=0x 2002000, [WRONG]

----- STREAM OFF -----

----- Ending -----
```

- Scan during streaming on


```

C:\Users\mtk31366\Desktop>python eye_scan_tool_streamon.py
adb is already running as root
Start EYE_SCAN
sensor_idx=1 drv_sensor_idx=0
sensor mode = 0
CDR_DELAY from 0 to 31
EQ_OFFSET from -31 to 31
/sys/devices/platform/soc/1a00e000.seninf-top

----- launch camera for init -----

----- STREAM OFF -----

----- CDR scan -----

[EYE_SCAN SUCCESS] set register:
save EYE_SCAN_KEYS_EQ_BW to seninf_ctx val=0x1

----- set parameters -----

[EYE_SCAN SUCCESS] set register:
save EYE_SCAN_KEYS_CDR_DELAY to seninf_ctx val=0x0
[EYE_SCAN SUCCESS] set register:
save EYE_SCAN_KEYS_EQ_OFFSET to seninf_ctx val=0x1f

----- launch camera -----

[EYE_SCAN SUCCESS] set register:
SENINF_WRITE_REG set CSIRX_MAC_CSI2_IRQ_STATUS 0xffffffff
SENINF_WRITE_REG set SENINF_CSI2_IRQ_STATUS 0xffffffff
[EYE_SCAN SUCCESS] set register:
CSIRX_MAC_CSI2_IRQ_STATUS=0x 21034c
SENINF_CSI2_IRQ_STATUS=0x 0
check CSI2_IRQ_STATUS WRONG
CDR_DELAY=0x0 EQ_OFFSET=-31 CSIRX_MAC_CSI2_IRQ_STATUS=0x 21034c, [WRONG]

----- STREAM OFF -----

----- set parameters -----

[EYE_SCAN SUCCESS] set register:
save EYE_SCAN_KEYS_CDR_DELAY to seninf_ctx val=0x1
[EYE_SCAN SUCCESS] set register:
save EYE_SCAN_KEYS_EQ_OFFSET to seninf_ctx val=0x1f

----- launch camera -----

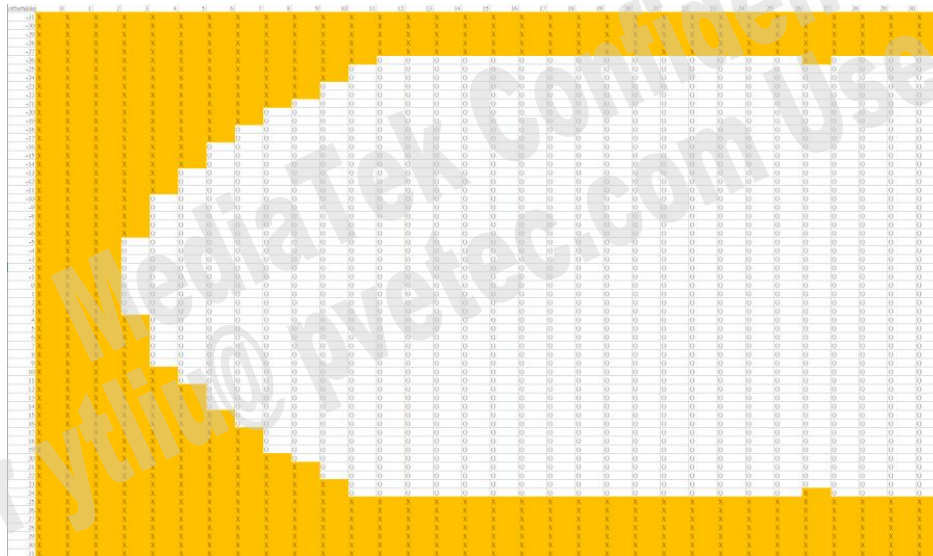
[EYE_SCAN SUCCESS] set register:
SENINF_WRITE_REG set CSIRX_MAC_CSI2_IRQ_STATUS 0xffffffff

```

- Result

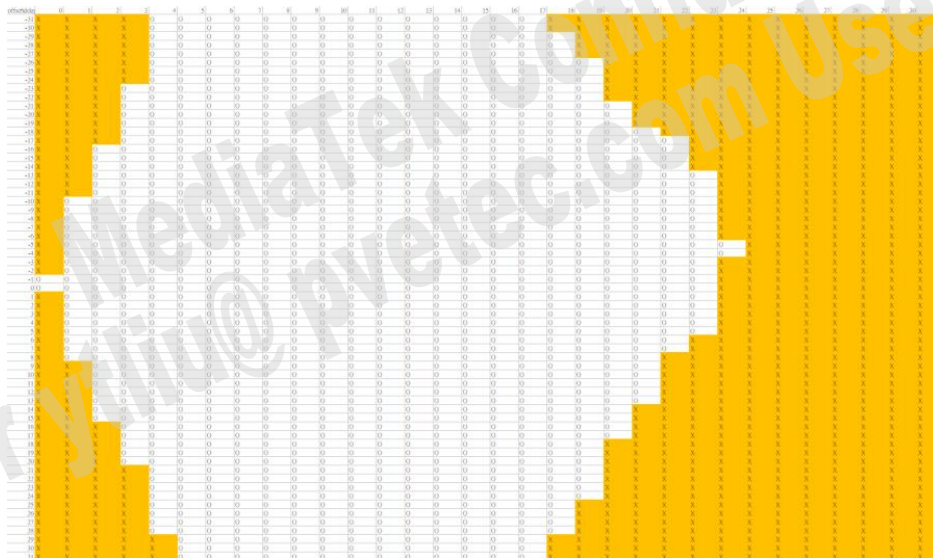
- Sensor: IMX766 (C-PHY)

- Mode 0
 - datarate (symbols per second) = (pixelrate * rawbit / 16 * 7) / trio = (462170000 * 10 / 16 * 7) / 3 = 673997916.6666
 - result.csv



– Sensor: IMX766 (C-PHY)

- mode 11
- datarate (symbols per second) = (pixelrate * rawbit / 16 * 7) / trio = (2122970000 * 10 / 16 * 7) / 3 = 3095997916.6666
- result.csv



– Sensor: S5KSP9 (D-PHY)

- mode 1
- datarate (bit per second) = (pixelrate * rawbit) / lane = (586000000 * 10) / 4 = 1465000000
- result.csv



4 ADB Command & Register Information.

The above Python script uses a sequence of ADB commands to read/write registers. The details of the ADB command and registers are provided.

4.1 How To Run EYE_SCAN Relative ADB Command

```
adb shell "echo EYE_SCAN dts_idx cmd_name val(option) > ...../debug_ops; cat ...../debug_ops"
```

The full path of debug_ops is different at DX-2 & DX-3 & DX-4.

- debug_ops path @ DX-2
 - /sys/devices/platform/1a00e000.seninf_top/debug_ops
- debug_ops path @ DX-3
 - /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops
- debug_ops path @ DX-3
 - /sys/devices/platform/soc/3a300000.seninf-top/seninf-top/debug_ops

4.1.1 Index & Streamon

- dts_idx and sensor_idx
 - The dts_idx is described in the device tree.
 - 0: Main cam
 - 1: Sub cam
 - 2: Main2 cam
 - 3: Sub2 cam
 - 4: Main3 cam
 - The relationship between dts_idx and sensor_idx (for SenTest):
 $sensor_idx = 2 \ll dts_idx$
- Before using ADB command to read/write registers on-line, such sensor should be streaming.
 - It is recommended to use SenTest to stream on sensor before running EYE_SCAN relative ADB command (the same as Python script).
 - Command: `adb root & adb shell sentest_v4l2 sensor_idx sensor_mode`
 - For example, sensor_idx = 1, sensor_mode = 0.

```

C:\Users\MTK31366>adb shell sentest_v4l2 1 0
[main]sizeof long : 4
[main]argc num:3
debug---[init_sensor_list] ioctl fail, line=300
[test_driver_cid]sensor_fd:5 seninf_fd:6
[test_driver_cid]ret:0
[test_set_ext][test test_set_ext]
[test_set_ext][V4L2_CID_MTK_STAGGER_AE_CTRL] ret:0
[test_set_ext][V4L2_CID_MTK_STAGGER_AE_CTRL] ret:0
[test_CID][test try_ext_ctrl] scenario (by try format)
[test_CID][V4L2_CID_VBLANK] ret:0 VB:9072
[test_CID][V4L2_CID_HBLANK] ret:0 HB:2614
[test_CID][V4L2_CID_MTK_SENSOR_PIXEL_RATE] ret:0 pixel rate:-1849435868
[test_CID][V4L2_CID_MTK_STAGGER_INFO] ret:0 count:1 order:
[test_CID] 0
[test_CID][V4L2_CID_MAX_EXP_TIME]ret 0 exposure 0, max_exp_time 8381952
[test_CID][V4L2_CID_MAX_EXP_TIME]ret 0 exposure 1, max_exp_time 8381952
[test_CID][V4L2_CID_MAX_EXP_TIME]ret 0 exposure 2, max_exp_time 8381952
[test_static_param][V4L2_CID_MTK_SENSOR_STATIC_PARAM] VB 9072 HB 2614 FPS 30 pixel rate -1849435868
[test_static_param][V4L2_CID_MTK_SENSOR_STATIC_PARAM] VB 9072 HB 2614 FPS 30 pixel rate -1849435868
[test_static_param][V4L2_CID_MTK_SENSOR_STATIC_PARAM] VB 7320 HB 4372 FPS 30 pixel rate -1849435868
[test_static_param][V4L2_CID_MTK_SENSOR_STATIC_PARAM] VB 968 HB 3754 FPS 120 pixel rate -1209253010
[test_static_param][V4L2_CID_MTK_SENSOR_STATIC_PARAM] VB 1448 HB 2110 FPS 240 pixel rate -1849435868
[test_static_param][V4L2_CID_MTK_SENSOR_STATIC_PARAM] VB 144 HB 2429 FPS 960 pixel rate -1209253010
[test_static_param][V4L2_CID_MTK_SENSOR_STATIC_PARAM] VB 696 HB 1696 FPS 480 pixel rate -1209253010
[test_static_param][V4L2_CID_MTK_SENSOR_STATIC_PARAM] VB 144 HB 4574 FPS 480 pixel rate -1849435868
[test_static_param][V4L2_CID_MTK_SENSOR_STATIC_PARAM] VB 224 HB 22692 FPS 30 pixel rate 2037942857

```

4.2 Command Name & Value

Notify that the full path of debug_ops is different at DX-2 & DX-3.

After sending the ABD command, you would get a string beginning with [EYE_SCAN SUCCESS], which means such registers have been written without any error. Otherwise, a string beginning with [EYE_SCAN FAIL] means there is something wrong (e.g., the sensor is not streaming)

CMD Name	Register Func.	Purpose	Value Range	CMD
EQ_DG0_EN	CTLE L1	Improve the quality of MIPI signal	{0x0, 0x1}	echo EYE_SCAN dts_idx EQ_DG0_EN 0x1 > /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops; cat /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops
GET_EQ_DG0_EN			Not need	echo EYE_SCAN dts_idx GET_EQ_DG0_EN > /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops; cat /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops
EQ_SR0			0~15	echo EYE_SCAN dts_idx EQ_SR0 val > /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops; cat /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops
GET_EQ_SR0			Not need	echo EYE_SCAN dts_idx GET_EQ_SR0 > /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops; cat /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops
EQ_DG1_EN	CTLE L2		{0x0, 0x1}	echo EYE_SCAN dts_idx EQ_DG1_EN 0x1 > /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops; cat /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops
GET_EQ_DG1_EN			Not need	echo EYE_SCAN dts_idx GET_EQ_DG1_EN > /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops; cat

CMD Name	Register Func.	Purpose	Value Range	CMD
				/sys/devices/platform/soc/1a00e000.seninf-top/debug_ops
EQ_SR1			0~15	echo EYE_SCAN dts_idx EQ_SR1 val > /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops; cat /sys/devices/platform/soc/1a00e000.seninf-top/debug_op
GET_EQ_SR1			Not need	echo EYE_SCAN dts_idx GET_EQ_SR1 > /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops; cat /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops
EQ_BW	Buffer bandwidth		{0x0, 0x1, 0x3}	echo EYE_SCAN dts_idx EQ_BW val > /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops; cat /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops
GET_EQ_BW			Not need	echo EYE_SCAN dts_idx GET_EQ_BW > /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops; cat /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops
FLUSH_CRC_STATUS	Check seninf IRQ status	Eye scan	Not need	echo EYE_SCAN dts_idx FLUSH_CRC_STATUS > /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops; cat /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops
CDR_DELAY	Scan width of eye		CPHY: 0~31 DPHY: 0~254	echo EYE_SCAN dts_idx CDR_DELAY val > /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops; cat /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops
GET_CDR_DELAY			Not need	echo EYE_SCAN dts_idx GET_CDR_DELAY > /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops; cat /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops
CDR_DELAY_DPHY_EN			Not need	echo EYE_SCAN dts_idx CDR_DELAY_DPHY_EN > /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops; cat /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops
EQ_OFFSET	Scan height of eye		-31 ~ 31	echo EYE_SCAN dts_idx EQ_OFFSET val > /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops; cat /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops
GET_EQ_OFFSET	Scan height of eye		Not need	echo EYE_SCAN dts_idx GET_EQ_OFFSET > /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops; cat /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops

4.3 Example

- Terminal 1: Stream on sensor

```
PS C:\Users\MTK31366> adb shell
k6989v1_64:/ # sentest_v4l2 1 0
[main]sizeof long : 4
[main]argc num:3
debug--[init_sensor_fail] ioctl fail, line=300
[test_driver_cid]sensor_id:5 seninf_id:6
[test_driver_cid]ret:0
[test_set_ext][test test set_ext]
[test_set_ext][V4L2_CID_MIR_STAGGER_AE_CTRL] ret:0
[test_set_ext][V4L2_CID_MIR_STAGGER_AE_CTRL] ret:0
[test_CID][test try_ext_ctrl] scenario (by try format)
[test_CID][V4L2_CID_VBLANK] ret:0 VB:9072
[test_CID][V4L2_CID_HBLANK] ret:0 HB:2614
[test_CID][V4L2_CID_MIR_SENSOR_PIXEL_RATE] ret:0 pixel rate:-1849435868
[test_CID][V4L2_CID_MIR_STAGGER_INFO] ret:0 count order:
[test_CID] 0
[test_CID][V4L2_CID_MAX_EXP_TIME]ret 0 exposure 0, max_exp_time 8381952
[test_CID][V4L2_CID_MAX_EXP_TIME]ret 0 exposure 1, max_exp_time 8381952
[test_CID][V4L2_CID_MAX_EXP_TIME]ret 0 exposure 2, max_exp_time 8381952
[test_static_param][V4L2_CID_MIR_SENSOR_STATIC_PARAM] VB 9072 HB 2614 FPS 30 pixel rate -1849435868
[test_static_param][V4L2_CID_MIR_SENSOR_STATIC_PARAM] VB 9072 HB 2614 FPS 30 pixel rate -1849435868
[test_static_param][V4L2_CID_MIR_SENSOR_STATIC_PARAM] VB 7320 HB 4372 FPS 30 pixel rate -1849435868
[test_static_param][V4L2_CID_MIR_SENSOR_STATIC_PARAM] VB 968 HB 3754 FPS 120 pixel rate -1209253010
[test_static_param][V4L2_CID_MIR_SENSOR_STATIC_PARAM] VB 1448 HB 3110 FPS 240 pixel rate -1849435868
```

- Terminal 2: Use EYE_SCAN ADB command

```
PS C:\Users\MTK31366> adb shell
k6989v1_64:/ # echo EYE_SCAN 0 EQ_DG0_EN 0x1 > /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops; cat /sys/devic>
[EYE_SCAN SUCCESS] set register:
SENINF_BITS set RG_CS10_CDPHY_EQ_DG0_EN, val=0x1
SENINF_BITS set RG_CS10_CDPHY_EQ_DG0_EN, val=0x1
k6989v1_64:/ # echo EYE_SCAN 0 GET_EQ_DG0_EN > /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops; cat /sys/devic>
[EYE_SCAN SUCCESS] set register:
SENINF_READ_BITS get RG_CS10_CDPHY_EQ_DG0_EN, val=0x1
SENINF_READ_BITS get RG_CS10_CDPHY_EQ_DG0_EN, val=0x1
```

- Use EYE_SCAN ADB command before sensor is streamed on.

Note that you must stream on the sensor at least one time before using the ADB command for V4L2 link setup.

```
C:\Users\mtk31366>adb shell "echo EYE_SCAN 0 CDR_DELAY 0 > /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops;cat /sys/devices/platform/soc/1a00e000.seninf-top/debug_ops"
[EYE_SCAN SUCCESS] set register:
save EYE_SCAN_KEYS_CDR_DELAY to seninf.ctx val=0x0
```

While the sensor is streamed off, the input parameter will be saved and only be written to register when the sensor is streamed on next time.

Once the input parameter is written, it will be cleared.

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